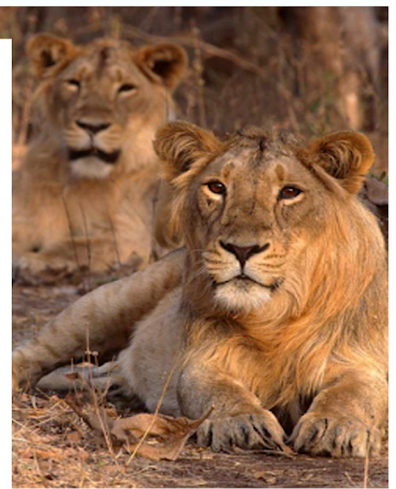
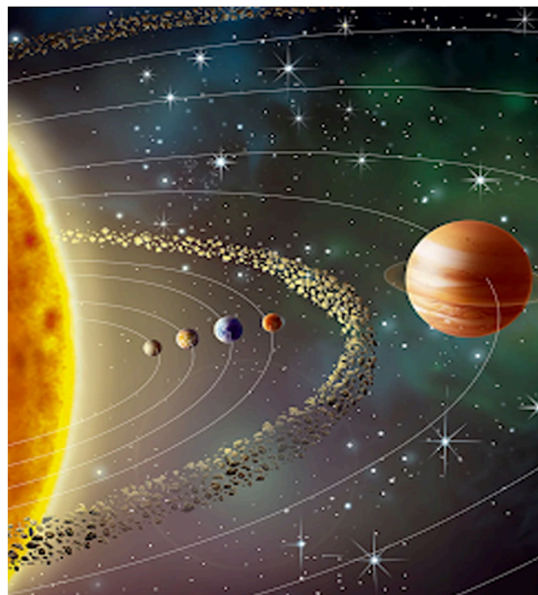


General Knowledge

World Geography

STUDY NOTES

Oceanography: Movements of Ocean Water



Oceanography: Movements of Ocean Water

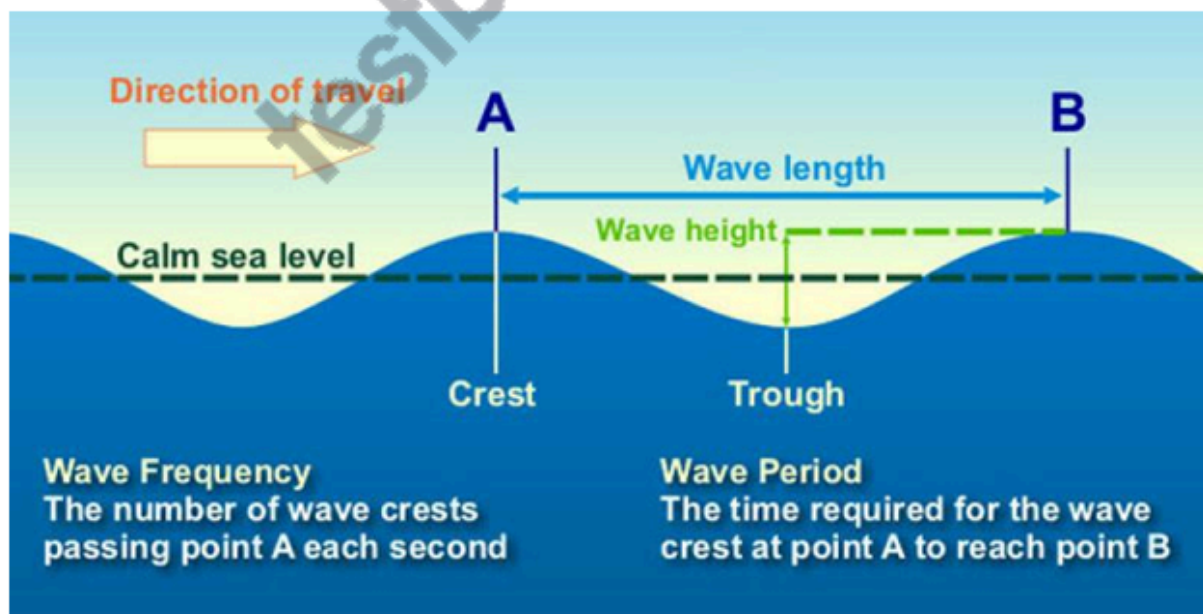
The ocean is a dynamic system influenced by various physical characteristics like temperature, salinity, and density, as well as external forces such as the sun, moon, and winds. These factors drive both horizontal and vertical movements in ocean water.

Horizontal motion refers to ocean currents and waves. Ocean currents are large, continuous flows of water moving in a specific direction, while waves represent the horizontal movement of water. In waves, the water itself doesn't travel; instead, it is the energy of the wave that moves across the surface.

Vertical motion involves the rise and fall of water levels in oceans and seas. This motion is primarily caused by the gravitational pull of the sun and moon, which causes the water to rise and fall twice a day in a process known as tides. Additionally, vertical movements include upwelling, where cold water rises from below the surface, and the sinking of warmer surface water. These movements also contribute to the vertical motion of ocean water.

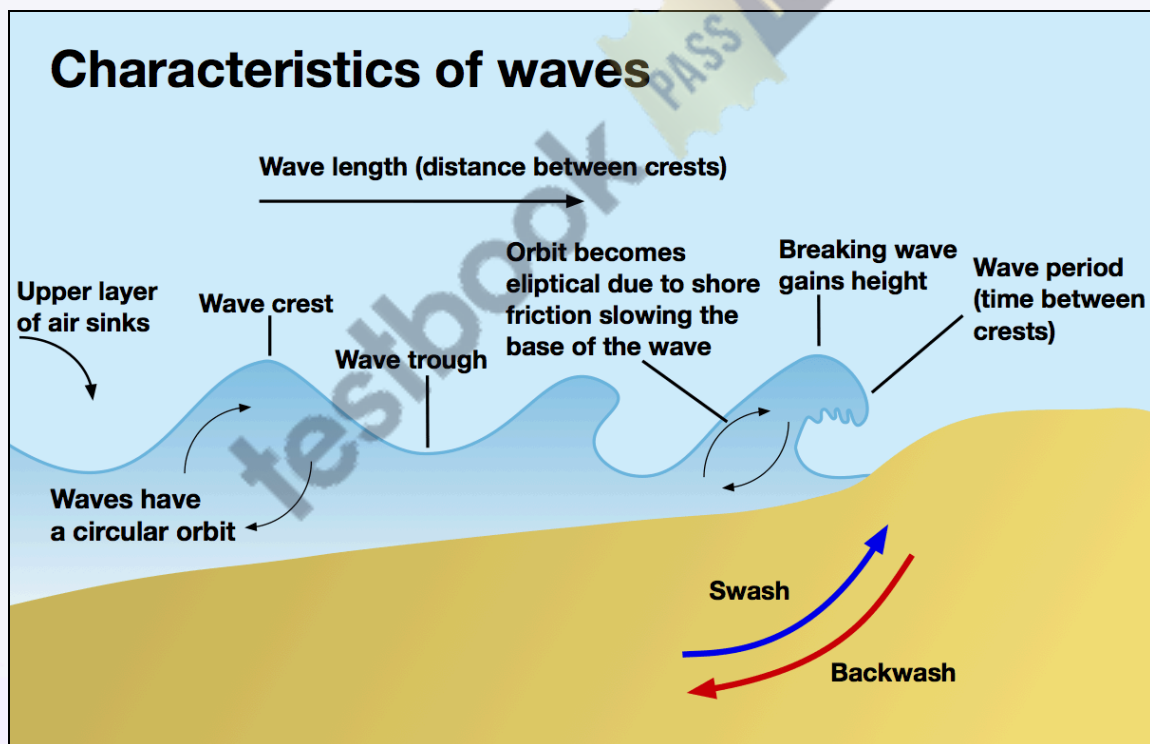
Waves

- **Waves are actually the energy**, not the water as such, which moves across the ocean surface.
- **Water particles only travel in a small circle as a wave passes.**



The anatomy of a wave

- **Wind provides energy to the waves.**
- Wind causes waves to travel in the ocean and the energy is released on shorelines.
- The motion of the surface water seldom affects the stagnant deep bottom water of the oceans.
- **As a wave approaches the beach, it slows down.** This is due to the friction occurring between the dynamic water and the sea floor. And, when the depth of water is less than half the wavelength of the wave, the wave breaks.
- The **largest waves are found in the open oceans.** Waves continue to grow larger as they move and absorb energy from the wind. Most of the waves are caused by the wind driving against water.
- When a breeze of two knots or less blows over calm water, small ripples form and grow as the wind speed increases until white caps appear in the breaking waves.
- Waves may travel thousands of km before rolling ashore, breaking and dissolving as surf.



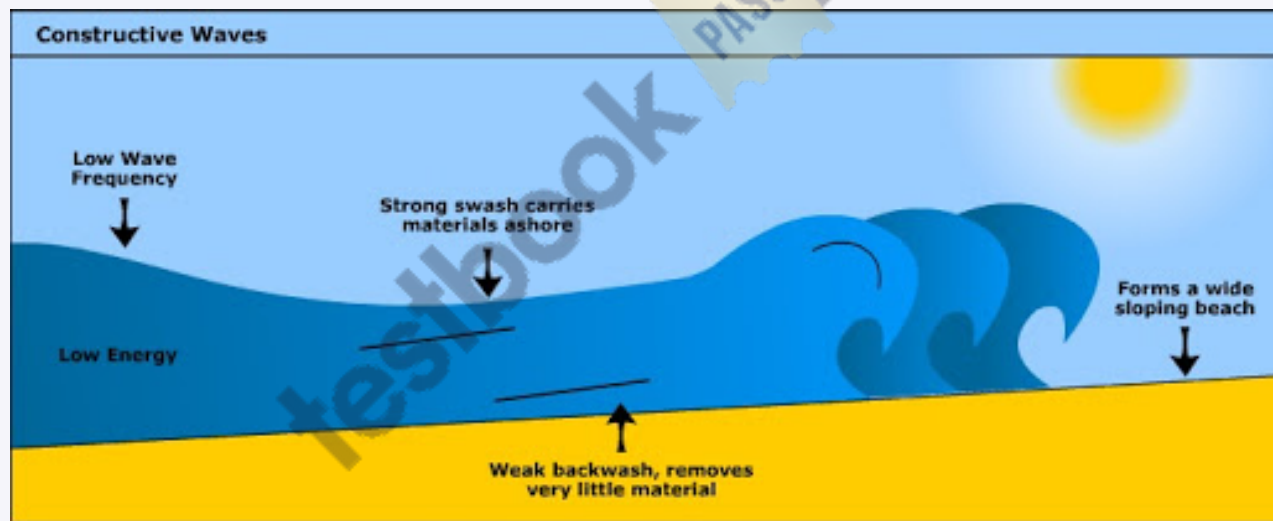
- A wave's **size and shape reveal its origin.**
- Steep waves are fairly young ones and are probably formed by local wind.
- Slow and steady waves originate from faraway places, possibly from another hemisphere.

- The maximum **wave height is determined by the strength of the wind**, i.e. how long it blows and the area over which it blows in a single direction.
- **Waves travel because the wind pushes the water body in its course while gravity pulls the crests of the waves downward.** The falling water pushes the former troughs upward, and the wave moves to a new position.

The actual **motion of the water beneath the waves is circular**. It indicates that things are carried up and forward as the wave approaches, and down and back as it passes.

Constructive waves -

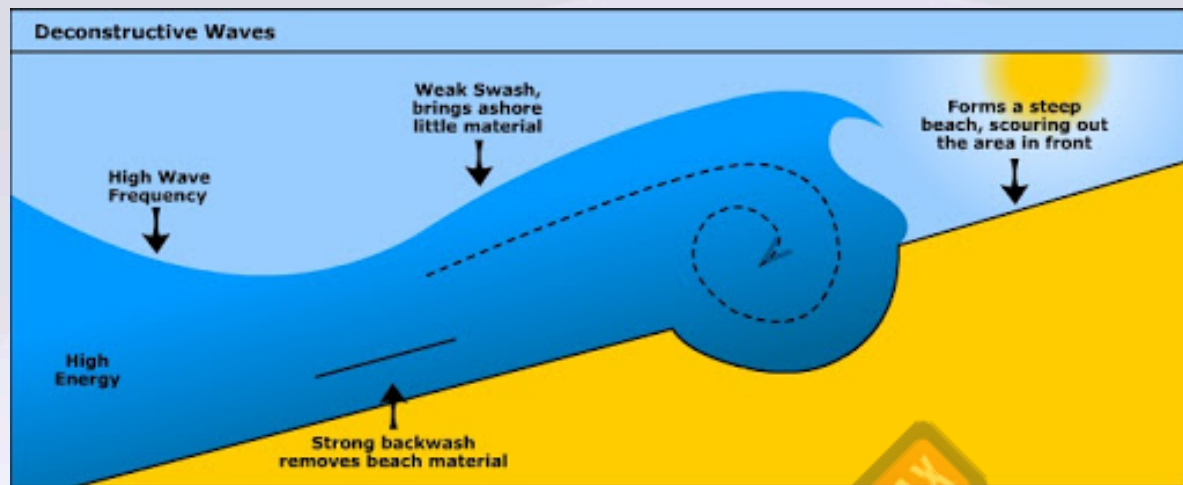
- They are created in calm weather and are less powerful than destructive waves.
- They break on the shore and deposit material, building up beaches.
- They have a swash that is stronger than the backwash.
- They have a long wavelength and are low in height.



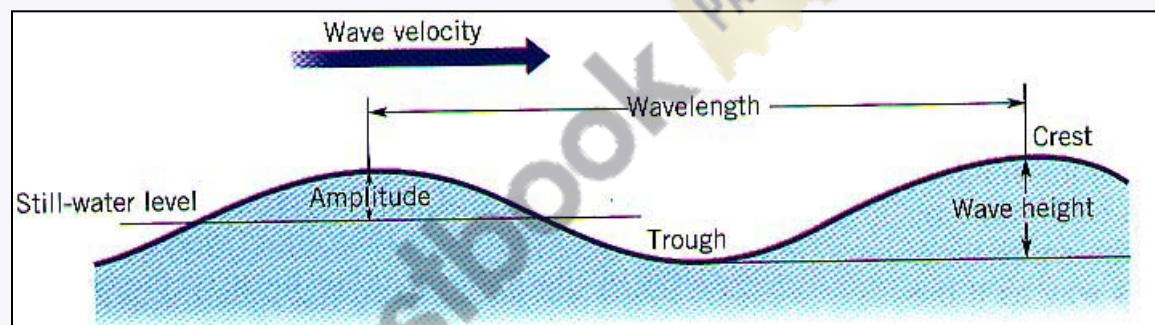
Deconstructive waves -

- Destructive waves are created in storm conditions.
- They are created from big, strong waves when the wind is powerful and has been blowing for a long time.
- They occur when wave energy is high and the wave has traveled over a long fetch.
- They tend to erode the coast.
- They have a stronger backwash than swash.

- They have a short wavelength and are high and steep.



Characteristics of Waves

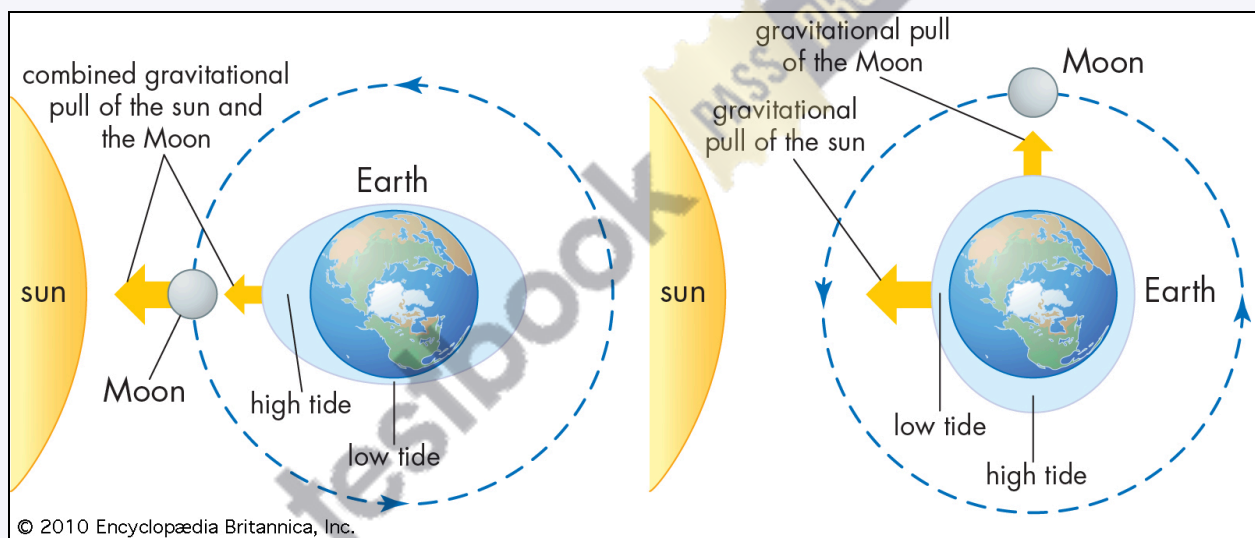


- **Wave crest and trough:** The highest and lowest points of a wave are called the crest and trough respectively.
- **Wave height:** It is the vertical distance from the bottom of a trough to the top of a crest of a wave.
- **Wave amplitude:** It is one-half of the wave height.
- **Wave period:** It is merely the time interval between two successive wave crests or troughs as they pass a fixed point.
- **Wavelength:** It is the horizontal distance between two successive crests.
- **Wave speed:** It is the rate at which the wave moves through the water, and is measured in knots.

- **Wave frequency:** It is the number of waves passing a given point during a one-second time interval.

Tides

- The periodical rise and fall of the sea level, once or twice a day, mainly due to the **attraction of the sun and the moon**, is called a tide.
- The movement of water caused by **meteorological effects** (winds and atmospheric pressure changes) is called **surges**.
- Surges are not regular like tides. The study of tides is very complex, spatially and temporally, as it has great variations in frequency, magnitude, and height.
- The moon's gravitational pull to a great extent and to a lesser extent the sun's gravitational pull, are the major causes for the occurrence of tides.

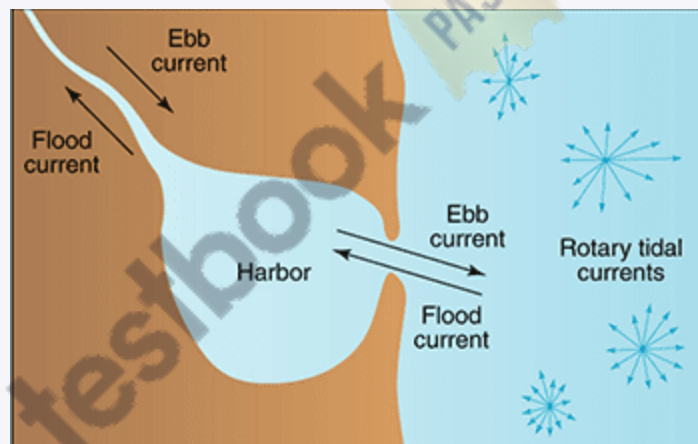


- Another factor is **centrifugal force**, which is the force that acts to counterbalance gravity. Together, the gravitational pull and the centrifugal force are responsible for creating the two major tidal bulges on the Earth.
- On the side of the earth facing the moon, a **tidal bulge** occurs while on the opposite side though the gravitational attraction of the moon is less as it is farther away, the centrifugal force causes a tidal bulge on the other side.
- The **'tide-generating' force** is the difference between these two forces; i.e. the gravitational attraction of the moon and the centrifugal force.

- On the surface of the earth, **nearest the moon, pull or the attractive force of the moon is greater than the centrifugal force**, and so there is a net force causing a **bulge towards the moon**.
- On the opposite side of the earth, the attractive force is less, as it is farther away from the moon, the centrifugal force is dominant. Hence, there is a net force away from the moon. It creates a second bulge away from the moon.
- On the surface of the earth, the horizontal tide-generating forces are more important than the vertical forces in generating the tidal bulges.
- The tidal bulges on wide continental shelves have greater height. When tidal bulges hit the mid-oceanic islands they become low.

Tidal currents:

- The shape of bays and estuaries along a coastline can also magnify the intensity of tides. Funnel-shaped bays greatly change tidal magnitudes. When the tide is channeled between islands or into bays and estuaries they are called **tidal currents**.



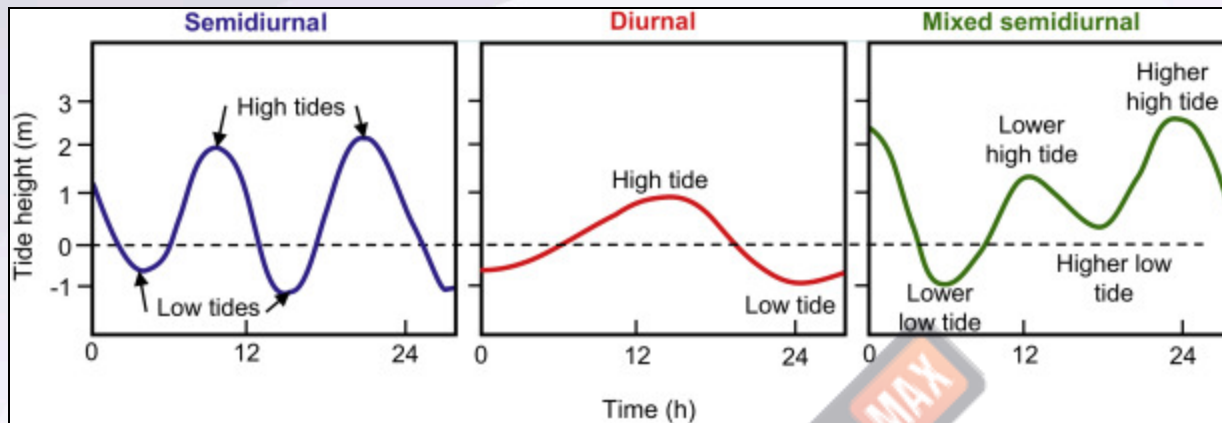
Types of Tides

- Tides vary in their **frequency, direction, and movement** from place to place and also from time to time.
- Tides may be grouped into various types based on their frequency of occurrence in one day or 24 hours or based on their height.

Tides based on Frequency

1. **Semi-diurnal tide:** The most common tidal pattern, featuring two high tides and two low tides each day. The successive high or low tides are approximately of the same height.

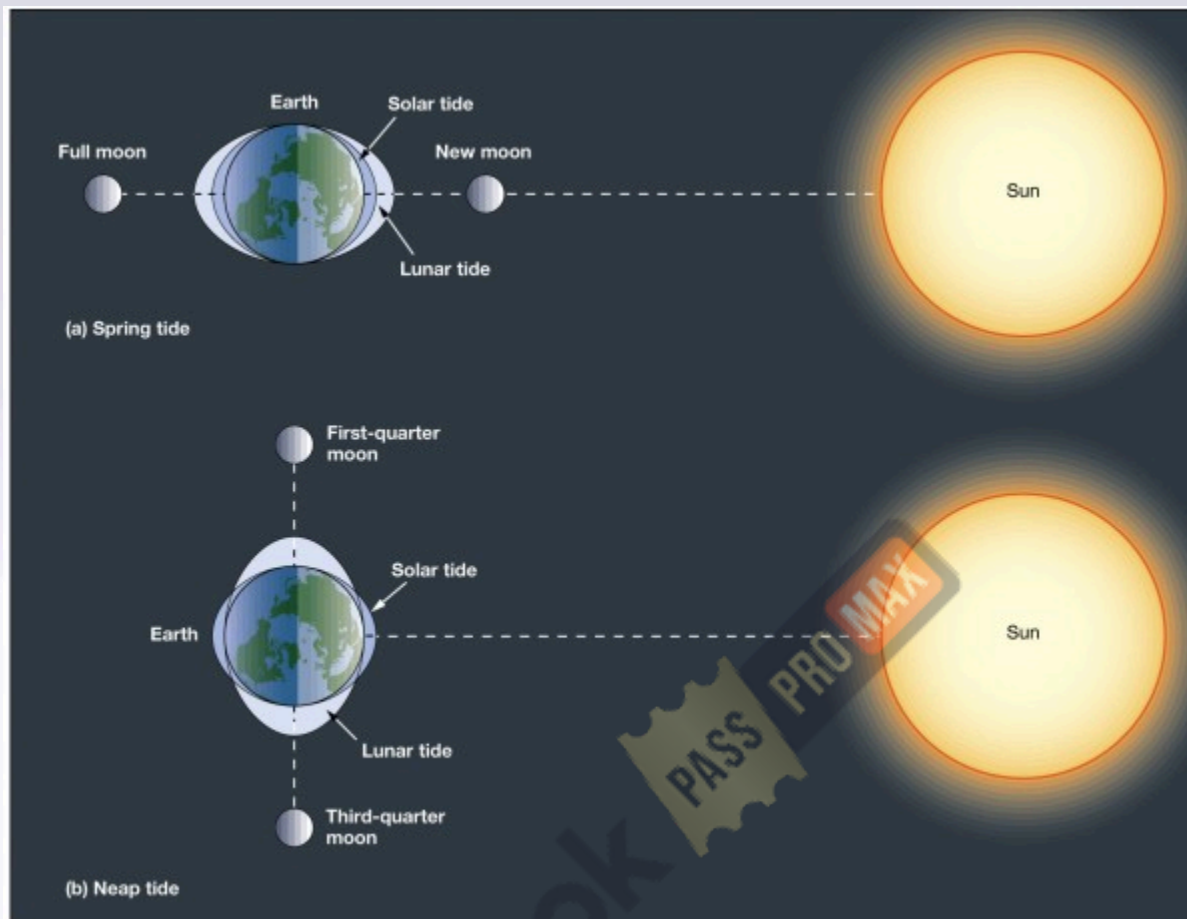
2. **Diurnal tide:** There is only one high tide and one low tide during each day. The successive high and low tides are approximately of the same height.
3. **Mixed tide:** Tides having variations in height are known as mixed tides. These tides generally occur along the west coast of North America and on many islands of the Pacific Ocean.



Tides based on the Sun, Moon, and the Earth Positions

The height of rising water (high tide) varies appreciably depending upon the position of the sun and moon with respect to the Earth. Spring tides and neap tides come under this category.

1. **Spring tides:** The position of both the sun and the moon in relation to the earth has direct bearing on tide height. When the sun, the moon, and the earth are in a straight line, the height of the tide will be higher. These are called spring tides and they occur twice a month, one during the full moon period and another during the new moon period.
2. **Neap tides:** Normally, there is a seven-day interval between the spring tides and neap tides. At this time the sun and moon are at right angles to each other and the forces of the sun and moon tend to counteract one another. The Moon's attraction, though more than twice as strong as the sun's, is diminished by the counteracting force of the sun's gravitational pull.



- Once a month, when the moon's orbit is closest to the earth (**perigee**), unusually high and low tides occur. During this time the tidal range is greater than normal.
- Two weeks later, when the moon is farthest from Earth (**apogee**), the moon's gravitational force is limited and the tidal ranges are less than their average heights.
- When the earth is closest to the sun (**perihelion**), around 3rd January each year, tidal ranges are also much greater, with unusually high and unusually low tides.
- When the earth is farthest from the sun (**aphelion**), around the 4th of July each year, tidal ranges are much less than average.
- The **time between the high tide and low tide**, when the water level is falling, is called the **ebb**.
- The **time between the low tide and high tide**, when the tide is rising, is called the **flow or flood**.

Importance of Tides

- Since tides are caused by the earth-moon-sun positions which are known accurately, the tides can be predicted well in advance. This helps the navigators and fishermen plan their activities. Tidal flows are of great importance in **navigation**.
- Tidal heights are very important, especially in harbors near rivers and within estuaries having shallow 'bars' at the entrance, **which prevent ships and boats from entering the harbor**.
- Tides are also helpful in **desilting the sediments** and in removing polluted water from river estuaries.
- Tides are used to generate **electrical power** (in Canada, France, Russia, and China). A 3 MW tidal power project at Durgaduani in Sunderbans of West Bengal is underway.